Operations

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Impossible mappings in HS

- Contradictory rankings / harmonic bounding (same as parallel OT)

(1)

/CV/	*COMPLEXONSET	DEP
☞ a. [.CV.]		
b. [.CCV.]	W 1	W 1

- No optimal path

  if /A/ → [C] via intermediate form: /A/ → B → C → [C]

  but /A/ → B is impossible, then /A/ $\not\to$ [C]

  all steps involve one operation application: can rule out /A/ → [C] by decomposing its operations

1 Coda-onset asymmetry

Generalization: deletion and assimilation in /…VC_1C_2V…/ *by default* targets C_1: […VC_2V…] *[…VC_1V…] (Wilson, 2000)

Diola Fogny (dyo, Niger-Congo, Gambia/Senegal)

(2) **Deletion in Diola Fogny** (Hopkins, 1995:20)

/na-ɾok-ɾok/	[naɾo_ɾok]	‘il a travaillé’ [he worked]
/ku-ɟam-ɔːɾ-ɟam-ɔːɾ/	[kuɟamɔː_ɟɔːɾ]	‘ils se sont entendus’ [they got along]
/let-a-ɟaw/	[lɛtaɟaw]	‘il n’ira pas’ [he won’t go]
/let-ku-ɟaw/	[lɛː_kuɟaw]	‘ils n’iront pas’ [they won’t go]

McCarthy’s (2008; 2019) analysis

Operations
Delete [place] node p → H
Delete placeless segments H → Ø

Constraints

- CodaCond: AOVFE [place] feature not linked to a syllable onset[1]
- HavePlace: AOVFE segment without a [place] feature
- Max(place): AOVFE deleted [place] feature
- Max(place): AOVFE deleted segment

Word-final consonants are onsets (Piggott, 1999)

(3) /let-ku-ɟaw/ → [.lɛː.ku.ɟa.w.]

/let-ku-ɟaw/	CodaCond	Max(place)	HavePlace	Max
a. [.let.ku.ɟa.w.]	W 1	L	L	
☞ b. [.leH.ku.ɟa.w.]		1	1	
c. [.let.Hu.ɟa.w.]	W 1	1	1	
d. [.leH.ku.ɟa.w.]			W 1	L
☞ e. [.le.ku.ɟa.w.]				1
☞ f. [.le.ku.ɟa.w.]				

There is no path to [.lɛː.tu.ɟa.w.]

- /t/ is a coda in an intermediate step
- /k/ → H does not improve on CodaCond
- imposing intermediate step rules out /k/ → Ø

Similar argument for place assimilation: /nb/ → Nb → [mb]

- Gen can only link [place] to placeless segments

These constraints do not work in parallel OT

[1] McCarthy (2008:279)

(4) /let-ku-ɟaw/ → [.lɛː.ku.ɟa.w.] in parallel OT

/let-ku-ɟaw/	CodaCond	Max(place)	HavePlace	Max
a. [.let.ku.ɟa.w.]	W 1	L		L
☞b. [.lɛː.ku.ɟa.w.]		1		1
☞c. [.lɛː.tu.ɟa.w.]		1		1

- no intermediate forms; whatever surfaces is an onset

difficult to formulate appropriate positional faithfulness constraints (Beckman, 1998) because

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Carib[2] (car, Cariban, Brazil/the Guianas/Venezuela)

(5) **Syncope in Carib** (Hoff, 1968:88, 125, 167)

[.kuː.mɨ.ɾɨ.]	'to be hungry'
[.ni.kuː.mɨ.ɾɨ.]	'he would be hungry'
[.kuː.mɨː.ja.]	'I am hungry'
[.kuː.mɨː.ke.pɨ.]	'to be no longer hungry'

- /kuːmɨɾɨ-ja/ → kuːmɨɾ_ja → kuːmɨ_ja
  both consonants are pre-vocalic in the input
  asymmetry only exists in intermediate form

caveat: only [ŋ, ŋ^{j}] surface word-finally (Hoff, 1968:55), raising possibility of epenthesis analysis (cf. Gildea, 1995)

2 Exceptions to the asymmetry

C_2 is targeted in /…VC_1C_2V…/ most often at stem-suffix boundaries (Lamont, 2015)

- *Place/affix (McCarthy, 2008:§3.7)

Nungon (yuw, Trans-New-Guinea, Papua New Guinea)

(6) **Assimilation in Nungon** (Sarvasy, 2017:105)

'gen'	'restrictive/dur'	
[.mak.kɔn.]	[.mak.gɔn.]	'mother'
[.mum.pɔn.]	[.mum.bɔn.]	'milk'
[.bin.tɔn.]	[.bin.dɔn.]	'skirt'
[.siŋ.kɔn.]	[.siŋ.gɔn.]	'falcon'
[.u.βa.hɔn.]	[.u.βa.ɣɔn.]	'pot'

[2] [.ka.ɽiʔ.n^{i}au.ɽaŋ.]

(7) /bin-gɔn/ → [.bin.dɔn.], Take 1

/bin-gɔn/	*Place/affix	CodaCond	Max(place)	HavePlace	*Link(place)	Max
a. [.bin.gɔn.]	W 2	2	L	L		
☹ b. [.bin.Hɔn.]	1	2	1	1		
💣 c. [.bin.gɔN.]	1	L 1	1	1		

- *Place/affix penalizes both consonants
- But final consonant isn't neutralized; there is contrast
  /-n/ '1.pl'
  /-ŋ/ '2/3.pl' (Sarvasy, 2017:233)

(8) /bin-gɔn/ → [.bin.dɔn.], Take 2

/bin-gɔn/	HavePlaceNasal	CodaCond	*Place/affix	Max(place)	HavePlace	Max	*Link(place)
a. [.bin.gɔn.]		2	W 2	L	L		
☞ b. [.bin.Hɔn.]		2	1	1	1		
c. [.biN.gɔn.]	W 1	L 1	W 2	1	1		
d. [.bin.gɔN.]	W 1	L 1	1	1	1		
e. [.bin.Hɔn.]		W 2	1		W 1		L
☞ f. [.bin.dɔn.]		1	1				1
g. [.bi.nɔn.]		1	1			W 1	L
☞ h. [.bin.dɔn.]		1	1				

- Weird: nasals more likely to assimilate (Jun, 1995)

but *Place/affix doesn't see clusters

(9) /uβa-gɔn/ → [.u.βa.gɔn.]

/uβa-gɔn/	HavePlaceNasal	CodaCond	*Place/affix	Max(place)	HavePlace	Max	*Link(place)
☹ a. [.u.βa.gɔn.]		1	2				
💣 b. [.u.βa.Hɔn.]		1	L 1	W 1	W 1		

- McCarthy (2008:298): well okay but these languages (Musey) don't allow [H] word internally
- ALIGN(H, L, ω, L) (McCarthy and Prince, 1993)?

  no, rules out .bin.Hin.
- …*VHV?

  no, richness of the base arguments

  */aqa/ $\not\rightarrow$ aHa → …
- …MAX(place)/V_V?

  same problem

so maybe debuccalization feeds [place] insertion

- but then why contrast [.u.βa.hɔn.] ~ [.u.βa.ɣɔn.]?
- and why not [.u.βa.dɔn.] (de Lacy, 2002, 2006)?

Afrikaans (afr, Indo European, South Africa)

(10) **Afrikaans diminutives** (Lamont, 2017)

[.pɑː.]	[.pɑː.ki.]	'father'
[.mɑn.]	[.mɑ.ni.ki.]	'man'
[.mɑːn.]	[.mɑːjŋ.ki.]	'moon'
[.rɑm.]	[.rɑ.mi.ki.]	'ram'
[.rɑːm.]	[.rɑːm.pi.]	'frame'
[.xə.sɑŋ.]	[.xə.sɑ.ŋi.ki.]	'religious song'
[.kuə.nəŋ.]	[kuə.nəŋ.ki.]	'king'

- bidirectional assimilation

  /nk/ → Nk → [ŋk]

  CODACOND ☺

  *PLACE/affix ☹

  /mk/ → mH → [mp]

  CODACOND ☹

  *PLACE/affix ☺

(11) A ranking paradox

winner ~ loser	CODACOND	*PLACE/affix
a. /mɑːn-ki/ → .mɑːN.ki. ~ .mɑn.Hi.	W	L
b. /rɑːm-ki/ → .rɑːm.Hi. ~ .rɑːN.ki.	L	W

not a problem in parallel – all $\widehat{\text{NC}}$ clusters satisfy CODACOND

- requires reference to surface form
- stringent faithfulness (de Lacy, 2002, 2006)
  FAITH{X}, FAITH{X, Y}, FAITH{X, Y, Z}

(12) /maːn-ki/ → [.maːjŋ.ki.] in parallel OT

/maːn-ki/	CODACOND	*LINK(place)	MAX{dor, lab, cor}	MAX{dor, lab}	*PLACE/affix	MAX{dor}
a. [.maːjn.ki.]	W 1	L	L		1	
☞ b. [.maːjŋ.ki.]		1	1		1	
c. [.maːjn.ti.]		1	1	W 1	L	W 1

(13) /raːm-ki/ → [.raːm.pi.] in parallel OT

/raːm-ki/	CODACOND	*LINK(place)	MAX{dor, lab, cor}	MAX{dor, lab}	*PLACE/affix	MAX{dor}
a. [.raːm.ki.]	W 1	L	L	L	W 1	L
b. [.raːjŋ.ki.]		1	1	1	W 1	L
☞ c. [.raːm.pi.]		1	1	1		1

What's missing in the HS analysis

- CODACOND is indirectly defined over clusters
- *PLACE/affix is not sensitive to clusters
- Clusters are the trigger of [place] assimilation
  AGREE(place): *[αplace][$\neg\alpha$place]
  Gradual assimilation is superfluous

(14) /maːn-ki/ → [.maːjŋ.ki.]

/maːn-ki/	AGREE(place)	HAVE(place)	*LINK(place)	MAX{dor, lab, cor}	MAX{dor, lab}	*PLACE/affix	MAX{dor}
a. [.maːn.ki.]	W 1	L		L		1	
☞b. [.maːN.ki.]		1		1		1	
c. [.maːn.Hi.]		1		1	W 1	L	W 1
d. [.maːN.ki.]		W 1	L			1	
☞e. [.maːŋ.ki.]			1			1	
☞f. [.maːŋ.ki.]						1	
g. [.maːŋ.Hi.]		W 1		W 1	W 1	L	W 1

(15) /raːm-ki/ → [.raːm.pi.]

/raːm-ki/	AGREE(place)	HAVE(place)	*LINK(place)	MAX{dor, lab, cor}	MAX{dor, lab}	*PLACE/affix	MAX{dor}
a. [.raːm.ki.]	W 1	L		L	L	W 1	L
b. [.raːN.ki.]		1		1	1	W 1	L
☞c. [.raːm.Hi.]		1		1	1		1
d. [.raːm.Hi.]		W 1	L				
☞e. [.raːm.pi.]			1				
☞f. [.raːm.pi.]							
g. [.raːN.pi.]		W 1		W 1	W 1		W 1

(16) /paː-ki/ → [.paː.ki.]

/paː-ki/	AGREE(place)	HAVE(place)	*LINK(place)	MAX{dor, lab, cor}	MAX{dor, lab}	*PLACE/affix	MAX{dor}
☞a. [.paː.ki.]						1	
b. [.paː.Hi.]		W 1		W 1	W 1	L	W 1

crucial cases are fed by syncope – require intermediate form

Chamicuro[3] (ccc, Arawakan, Peru)

[3] [.ˌt͡ʃa.me.ˈko.d͡lʒo.]

(17) **Nasal place assimilation** (Parker, 1988:8, 78, 107)

[.u.ˈne.ni.]	'I come'
[.pi.ˈne.ni.]	'you come'
[.i.ˈneŋ. .ˈka.na.]	'they come'

(18) **Syncope** (Parker, 1988:39, 73)

[.ke.ˈni.t͡ʃi.]	'worm-ABS'
[.piʔ.ˈto.t͡ʃi.]	'canoe-ABS'
[.me.ˈnut͡ʃ. .ˈka.na.]	'the tongues-ABS'
[.a.wa.ˈnat͡ʃ. .ˈka.na.]	'the mouths-ABS'

(19) **Syncope and place assimilation** (Parker, 1988:41)

[.u.ˈt͡ʃo.ni.]	'I walk'
[.u.ˈt͡ʃoŋ. .ˈka.ti.]	'I walked'
[.u.ˈkeʔ.mi.]	'I hear'
[.u.ˈkem. .ˈka.ti.]	'I heard'
[.ˈwus.mi.]	'I sing'
[.ˈwus.m̩. .ˈka.ti.]	'I sang'

- not a perfect example because manner asymmetry

main point: operations, constraints mutually dependent

3 What about deletion?

(20) **Deletion in Diola Fogny** (Hopkins, 1995:20)

/na-ɾok-ɾok/	[naɾo_ɾok]	'il a travaillé' [he worked]
/ku-ɟam-ɔːɾ-ɟam-ɔːɾ/	[kuɟamɔː_ɟɔːɾ]	'ils se sont entendus' [they got along]
/let-a-ɟaw/	[lɛtaɟaw]	'il n'ira pas' [he won't go]
/let-ku-ɟaw/	[lɛː_kuɟaw]	'ils n'iront pas' [they won't go]

If gradual, intermediate form has a coda

- /na-ɾok-ɾok/ → .na.ɾoH.ɾok. → .na.ɾo.ɾok.

If not, same problem as parallel OT

- /na-ɾok-ɾok/ → .na.ɾo.ɾok. ∼ .na.ɾo.kok.

Unless syllabification is gradual (Elfner, 2009; Torres-Tamarit, 2012)...

- /na-ɾok-ɾok/ → (na) ɾokɾok → ··· → (na)(ɾok)(ɾok) → (na)(ɾo_)(ɾok) ¬ (na)(ɾo)(k_ok)

4 Gradual prosodification

McCarthy (2008) allows syllabification to allow freely in parallel with other operations

- Elfner (2009) and Torres-Tamarit (2012) argue that syllabification is an operation on equal footing

Operations (following Elfner, 2009)

- Core syllabification

  XY $\rightarrow$ $(\overset{\mu}{\text{XY}})$

  onset+nucleus

- Syllable projection

  X $\rightarrow$ (X) or $(\overset{\mu}{\text{X}})$

  minor syllable or onsetless syllable

- Adjunction

  X(…) $\rightarrow$ (X…)

  complex onset

  (…)X $\rightarrow$ (…X) or $(\ldots\overset{\mu}{\text{X}})$

  (complex) coda

Core syllabification looks decomposable

- XY $\rightarrow$ $\text{X}(\overset{\mu}{\text{Y}})$ $\rightarrow$ $(\overset{\mu}{\text{XY}})$

but it is necessary to force /VCV/ to surface as [(V)(CV)] not [(VC)(V)] (default outcome / universal) (Elfner, 2009:9-11)

Constraints

- ONSET:[4] AOVFE onsetless syllable
- *CODA:[5] AOVFE closed syllable
- PARSE(seg):[6] AOVFE segment not in a syllable
- HD(σ):[7] AOVFE syllable without a mora
- ALL-σ-L:[8] For every syllable in a word, assign one violation for every segment that separates it from the left edge of the word.

[4] (Prince and Smolensky, 1993/2004:20)

[5] (Prince and Smolensky, 1993/2004:41)

[6] (McCarthy and Prince, 1994:344)

[7] (Selkirk, 1996:190)

[8] (Mester and Padgett, 1994:80)

(21) Bad prediction w/o core syllabification (assume μ iff V)

/CVCV/	PARSE(seg)	HD(σ)	ONSET	ALL-σ-L	*CODA
a. CVCV	W 4		L	L	
b. (C)VCV	3	W 1	L	L	
☞ c. C$^{\mu}$(V)CV	3		1	1	
d. CVC$^{\mu}$(V)	3		1	W 3	
e. C$^{\mu}$(V)CV	W 3		W 1	W 1	
☞ f. $^{\mu}$(CV)CV	2				
g. C$^{\mu}$(VC)V	2		W 1	W 1	W 1
h. C$^{\mu}$(V)C$^{\mu}$(V)	2		W 2	W 4	
i. $^{\mu}$(CV)CV	W 2			L	L
☠ j. $^{\mu}$(CVC)V	1			1	1
k. $^{\mu}$(CV)C$^{\mu}$(V)	1		W 1	W 3	L

- is it a problem? depends on how resyllabification works

(22) Impossible with core syllabification (assume μ iff V)

/CVCV/	PARSE(seg)	ALL-σ-L	HD(σ)	ONSET	*CODA
a. CVCV	W 4				
☞ b. $^{\mu}$(CV)CV	2				
c. CV$^{\mu}$(CV)	2	W 2			
d. $^{\mu}$(CV)CV	W 2	L			
☞ e. $^{\mu}$(CV)$^{\mu}$(CV)		2			
f. $^{\mu}$(CVC)V	W 1	L			W 1

- this is the default
  can be overridden by stress etc.

Gradual syllabification obviates gradual deletion[9]

CODACOND: AOVFE [place] feature not linked to an onset

[9] joint work with Ruby Noble

- (ba$\overset{*}{\text{n}}$)(ka)
- (ba)$\overset{*}{\text{n}}$(ka)

  ⇒ syllabification *also* improves on CodaCond

(23) Coda-onset asymmetry via gradual syllabification

/let-ku-ɟaw/	Parse(seg)	CodaCond	*ComplexOnset	Max	Max(place)
a. letkuɟaw	W 8	W 5			
☞ b. (le)tkuɟaw	6	4			
☞ c. let(ku)ɟaw	6	4			
☞ d. letku(ɟa)w	6	4			
☜ e. let_uɟaw	W 7	4		W 1	W 1
⋮					
f. (le)t(ku)(ɟa)(w_)	W 1	W 1		L	L
g. (le)H(ku)(ɟa)(w_)	W 1			L	1
h. (let)(ku)(ɟa)(w_)		W 1		L	L
i. (le)(tku)(ɟa)(w_)			W 1	L	L
☞ j. (le)(ku)(ɟa)(w_)				1	1
☞ k. [(le)(ku)(ɟa)(w_)]					

- Core syllabification harmonically bounds deletion

✔ /CVC_1C_2V/ → *[CVC_1_V] is impossible: *CVC_1_V

one formal consequence: if CodaCond is active in a language, stray consonants that survive are onsets of minor syllables, not unsyllabified

✔ final consonant is definitely an onset

Does this work for other gradual deletions discussed by McCarthy (2019)? idk

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Dakota[10] (dak, Siouan-Catawban, USA/Canada)

[10] [.da.ˈkʰod.ʔi.a.pi.]

(24) **Dakota stress** (Shaw, 1985:175)

a. [.wa.ˈkte.] ‘I kill’

b. [.ma.ˈja.kte.] ‘you kill me’

c. [.wi.ˈt͡ʃa.ja.kte.] ‘you kill them’

d. [.o.ˈwi.t͡ʃa.ja.kte.] ‘you kill them there’

e. [.ˈkte.] ‘he/she/it kills’

- I’m assuming onset maximization because the syllable boundaries aren’t marked in the source

(25) **Epenthetic vowels are invisible to stress** (Shaw, 1985:184)

a. /t͡ʃek/ [.ˈt͡ʃe.ka.] ‘stagger’

b. /t͡ʃap/ [.ˈt͡ʃa.pa.] ‘trot’

c. /t’ĩs/ [.ˈt’ĩ.za.] ‘draw tight’

Elfner’s (2009) analysis

- *Coda $\gg$ Hd(σ) forces final C into minor syllables

  /t͡ʃek/ → (t͡ʃ$\overset{\mu}{\text{e}}$)k → (t͡ʃ$\overset{\mu}{\text{e}}$)(k)

- the minor syllable repels stress (Cho and King, 2003; Saba Kirchner, 2010)

  → (ˈt͡ʃ$\overset{\mu}{\text{e}}$)(k) *(t͡ʃ$\overset{\mu}{\text{e}}$)ˈ(k)

- Hd(σ) $\gg$ Dep motivates vowel epenthesis

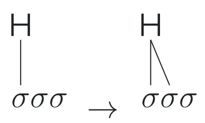
  → (ˈt͡ʃ$\overset{\mu}{\text{e}}$)(k$\overset{\mu}{\text{a}}$)

  too late for stress to land on it

parallel OT approach: Dep/stressed (Alderete, 1999:36)

Another *compound* operation

Uncontroversial: tonal spreading is an operation (McCarthy, 2006, 2007; McCarthy et al., 2012)

- 

- violates Dep(link)

Uncontroversial: epenthesis is an operation

- (ˈt͡ʃ$\overset{\mu}{\text{e}}$)(k) → (ˈt͡ʃ$\overset{\mu}{\text{e}}$)(k$\overset{\mu}{\text{a}}$)

- Lin (2014) argues for a gradual path à la deletion
  e.g., Ø → ə → a

Arapaho[11] (arp, Algic, USA)

[11] [.hɪ.nɔ́.nɔ.ʔɛ.tí́ːt.]

(26) **Tone-driven epenthesis in Arapaho** (Gleim, 2019:1)

a. /hoːw-bén/ [.hoːw.bén.] 's/he doesn't drink'

b. /hoːw-Ⓗbetéeː/ [.hoː.wú.be.té.eː.] 's/he doesn't dance'

- I'm making an assumption about how to syllabify *éee*
- Ⓗ violates *Float, needs a host to link to
- inserting a vowel *and* linking it works
  /Ⓗ/ → ú
- just inserting a vowel does not
  /Ⓗ/ → u Ⓗ
  violates Dep without improving on phonotactics
  introduces a lookahead problem

5 Footing

Waorani[12] (auc, isolate, Ecuador/Peru)

[12] [.wao. .te.ɾe.ɾo.]

(27) **Unsuffixed stems** (Lester, 1994:18-20)

a. $(\acute{\sigma})$ ˈbe 'drink'

b. $(\acute{\sigma}\sigma)$ ˈmõ.ĩ 'bed'

c. $(\grave{\sigma})(\acute{\sigma}\sigma)$ ˌdĩ.ˌka.gõ 'ear ornament'

d. $(\grave{\sigma}\sigma)(\acute{\sigma}\sigma)$ ˌõ.mĩŋ.ˈka.pæ̃ 'rapids'

e. $(\grave{\sigma}\sigma)(\grave{\sigma})(\acute{\sigma}\sigma)$ ˌõ.nõ.ˌŋẽ.ˈne.pæ̃ 'wing'

f. $(\grave{\sigma}\sigma)(\grave{\sigma}\sigma)(\acute{\sigma}\sigma)$ ˌõ.nõŋ.ˌka.de.ˈmẽ.ŋõ 'navel'

- bidirectional
  $\ldots\sigma\sigma\ltimes \rightarrow \ldots(\acute{\sigma}\sigma)\ltimes$
  $\rtimes\sigma\sigma\sigma\sigma\ldots \rightarrow \rtimes(\grave{\sigma}\sigma)\sigma\sigma\ldots \rightarrow \rtimes(\grave{\sigma}\sigma)(\grave{\sigma}\sigma)\ldots$

(28) **Monosyllabic suffix strings** (Lester, 1994:103-115)

a. $(\acute{\sigma}-\sigma)$ ˈpõ−mi 'you come'

b. $(\grave{\sigma})(\acute{\sigma}-\sigma)$ ˌwæ̃.ˈnõ−mo 'I kill'

c. $(\grave{\sigma}\sigma)(\acute{\sigma}-\sigma)$ ˌje.wæ̃.ˈmõ−ŋã 'he writes (carves)'

d. $(\grave{\sigma}\sigma)(\grave{\sigma})(\acute{\sigma}-\sigma)$ ˌmã.nõ.ˌmõ.ˈka−kã 'he is making ear holes (pierced)'

e. $(\grave{\sigma}\sigma)(\grave{\sigma}\sigma)(\acute{\sigma}-\sigma)$ ˌgã.po.ˌgẽ.ne.ˈwa−bo 'he touches tongue'

(29) **Disyllabic suffix strings** (a-c, e: Lester, 1994:103-115; d: Pike, 1964:426)

a. $(\grave{\sigma}) - (\acute{\sigma}\sigma)$ ˌpo–ˈmɪ.pa ‘you come’

b. $(\grave{\sigma}\sigma) - (\acute{\sigma}\sigma)$ ˌwæ̃.nõ–ˈmo.ni ‘we kill’

c. $(\grave{\sigma}\sigma)(\grave{\sigma}) - (\acute{\sigma}\sigma)$ ˌa.pæ̃.ˌne–ˈbo.pa ‘I speak’

d. $(\grave{\sigma}\sigma)(\grave{\sigma}\sigma) - (\acute{\sigma}\sigma)$ ˌpæ.dæ.ˌpõ.nõ–ˈɲãm.ba ‘he handed it over’

e. $(\grave{\sigma}\sigma)(\grave{\sigma}\sigma)(\grave{\sigma}) - (\acute{\sigma}\sigma)$ ˌgã.po.ˌgẽ.ne.ˌwa–ˈbo.pa ‘he touches tongue’

(30) **Trisyllabic suffix strings** (Lester, 1994:104-115)

a. $(\grave{\sigma} - \sigma)(\acute{\sigma}\sigma)$ ˌpõ–ta.ˈbo.pa ‘I came’

b. $(\grave{\sigma})(\grave{\sigma} - \sigma)(\acute{\sigma}\sigma)$ ˌda.ˌdõ–ta.ˈbo.pa ‘I fished’

c. $(\grave{\sigma}\sigma)(\grave{\sigma} - \sigma)(\acute{\sigma}\sigma)$ ˌkæ.ka.ˌpo–ta.ˈbo.pa ‘my knee hurt’

d. $(\grave{\sigma}\sigma)(\grave{\sigma})(\grave{\sigma} - \sigma)(\acute{\sigma}\sigma)$ ˌmã.nõ.ˌmõ.ˌka–dã.ˈnɪ.pa ‘they make ear holes (pierced)’

e. $(\grave{\sigma}\sigma)(\grave{\sigma}\sigma)(\grave{\sigma} - \sigma)(\acute{\sigma}\sigma)$ ˌgã.po.ˌgẽ.ne.ˌwa–ta.ˈbo.pa ‘he touched (the) tongue’

(31) **Tetrasyllabic suffix strings** (Lester, 1994:105-116)

a. $(\grave{\sigma}) - (\grave{\sigma}\sigma)(\acute{\sigma}\sigma)$ ˌpõ–ˌta.mõ.ˈna.pa ‘we (two) came’

b. $(\grave{\sigma}\sigma) - (\grave{\sigma}\sigma)(\acute{\sigma}\sigma)$ ˌwæ̃.æ̃–ˌta.mõ.ˈna.pa ‘we (two) fell down’

c. $(\grave{\sigma}\sigma)(\grave{\sigma}) - (\grave{\sigma}\sigma)(\acute{\sigma}\sigma)$ ˌje.wæ̃.ˌmõ–ˌta.mõ.ˈna.pa ‘we (two) wrote (carved)’

d. $(\grave{\sigma}\sigma)(\grave{\sigma}\sigma) - (\grave{\sigma}\sigma)(\acute{\sigma}\sigma)$ ˌmã.nõ.ˈmõ.ka–ˌta.mõ.ˈna.pa ‘we (two) made ear holes’

e. $(\grave{\sigma}\sigma)(\grave{\sigma}\sigma)(\grave{\sigma}) - (\grave{\sigma}\sigma)(\acute{\sigma}\sigma)$ ˌgã.po.ˌgẽ.ne.ˌwa–ta.mõ.ˈna.pa ‘we (two) touched (the) tongue’

(32) **Pentasyllabic suffix strings** (Lester, 1994:105-116)

a. $(\grave{\sigma}) - (\grave{\sigma}\sigma)\sigma(\acute{\sigma}\sigma)$ ˌpõ–ˌkæ.dõ.mõ.ˈnaĩ.pa ‘we (two) would have come’

b. $(\grave{\sigma}\sigma) - (\grave{\sigma}\sigma)\sigma(\acute{\sigma}\sigma)$ ˌwæ̃.æ̃–ˌkæ.dõ.mõ.ˈnaĩ.pa ‘we (two) would have fallen down’

c. $(\grave{\sigma}\sigma)(\grave{\sigma}) - (\grave{\sigma}\sigma)\sigma(\acute{\sigma}\sigma)$ ˌje.wæ̃.ˌmõ–ˌkæ.dõ.mõ.ˈnaĩ.pa ‘we (two) would have written (carved)’

d. $(\grave{\sigma}\sigma)(\grave{\sigma}\sigma) - (\grave{\sigma}\sigma)\sigma(\acute{\sigma}\sigma)$ ˌmã.nõ.ˌmõ.ka–ˌkæ.dõ.mõ.ˈnaĩ.pa ‘we (two) would have made ear holes’

e. $(\grave{\sigma}\sigma)(\grave{\sigma}\sigma)(\grave{\sigma}) - (\grave{\sigma}\sigma)\sigma(\acute{\sigma}\sigma)$ ˌgã.po.ˌgẽ.ne.ˌwa–ˌkæ.dõ.mõ.ˈnaĩ.pa ‘we (two) would have touched (the) tongue’

- cyclic generalization
  suffix strings parsed before roots

Operations following Pruitt (2010, 2012)

- $\sigma\sigma \rightarrow (\acute{\sigma}\sigma)$ or $(\sigma\acute{\sigma})$
  trochee or iamb

- $\sigma \rightarrow (\acute{\sigma})$

  monosyllabic foot

  I like to call these hooves, but this isn't standard

Constraints

- PARSE(σ):[13] AOVFE syllable not in a foot
- TROCHEE:[14] AOVFE right-headed foot
- IAMB: AOVFE left-headed foot
- ALL-FEET-LEFT:[15] For every foot in a word, assign one violation for every syllable that intervenes between the left edge of the foot and the left edge of the word.
- ALL-FEET-RIGHT: For every foot in a word, assign one violation for every syllable that intervenes between the right edge of the foot and the right edge of the word.
- FOOTLEFT:[16] Assign one violation for every word without a foot at its left edge.

[13] (Prince and Smolensky, 1993/2004:69)

[14] (Lamont, 2023)

[15] (McCarthy and Prince, 1993:80)

[16] (Kager, 1999:169)

(33) /õnõŋẽnepæ̃/ → [ˌõ.nõ.ˌŋẽ.ˈne.pæ̃]

/õnõŋẽnepæ̃/	PARSE(σ)	TROCHEE	IAMB	FOOTRIGHT	ALL-FEET-LEFT	ALL-FEET-RIGHT
⋮						
a. .õ.nõ.ŋẽ.ne.pæ̃.	W 5		L	W 1	L	
b. (.ˈõ.nõ.).ŋẽ.ne.pæ̃.	3		1	W 1	L	W 3
☞c. .õ.nõ.ŋẽ.(.ˈne.pæ̃.)	3		1		3	
d. .õ.nõ.ŋẽ.(.ne.ˈpæ̃.)	3	W 1	L		3	
e. .õ.nõ.ŋẽ.ne.(.ˈpæ̃.)	W 4	W 1	W 1		W 4	
f. .õ.nõ.ŋẽ.(.ˈne.pæ̃.)	W 3		L 1		3	L
☞g. (.ˌõ.nõ.).ŋẽ.(.ˈne.pæ̃.)	1		2		3	3
h. .õ.(.ˌnõ.ŋẽ.)(.ˈne.pæ̃.)	1		2		W 4	L 2
i. (.ˌõ.nõ.).ŋẽ.(.ˈne.pæ̃.)	W 1	L	L 2		L 3	L 3
☞j. (.ˌõ.nõ.)(.ˌŋẽ.)(.ˈne.pæ̃.)		1	3		5	5

can't do this in parallel OT (Hyde, 2008, 2012a,b, 2016)

(34) Hooves are attracted to word edges

/õnõŋẽnepæ̃/	PARSE(σ)	TROCHEE	IAMB	FOOTRIGHT	ALL-FEET-LEFT	ALL-FEET-RIGHT
a. .õ.nõ.ŋẽ.ne.pæ̃.	W 5	L	L	W 1	L	L
☹ b. (.ˌõ.nõ.)(.ˌŋẽ.)(.ˈne.pæ̃.)		1	3		5	5
c. (.ˌõ.)(.ˌnõ.ŋẽ.)(.ˈne.pæ̃.)		1	3		L 4	W 6
d. (.ˌõ.nõ.)(.ˌŋẽ.ne.)(.ˈpæ̃.)		1	3		W 6	L 4

- Fix an order $F_1 \ldots F_n F_{n+1} \ldots$
- F_{n+1} wants to be left-aligned, prefers F_n to be a hoof
- F_n likewise wants F_{n-1} to be a hoof, etc.

5.1 Parse constraints

(35) Problem: PARSE(σ) ≫ PARSE(seg) predicts no syllables

/ba/	PARSE(σ)	PARSE(seg)
💣 a. ba	L	W 2
☹ b. .ba.	1	

(36) Solution: pseudo stringent constraints

/ba/	PARSE(seg)/{σ}	PARSE(seg)/{F}
a. ba	W 2	2
☞ b. .ba.		2
c. .ba.		W 2
☞ d. (.ˈba.)		

necessary in HS; my intuition is that it would eliminate a lot of constraints on prosodic hierarchy. tbd

5.2 Alignment constraints are bad

Alignment (McCarthy and Prince, 1993; McCarthy, 2003; Hyde, 2012a)

- Parse(σ) $\gg$ All-Feet-Left $\gg$ All-Feet-Right
  $\rtimes\sigma\sigma\sigma\sigma\sigma\sigma \ldots \rightarrow \rtimes(\sigma\sigma)\sigma\sigma\sigma \ldots \rightarrow \rtimes(\sigma\sigma)(\sigma\sigma)\sigma \ldots \rightarrow \ldots$
- Parse(σ) $\gg$ All-Feet-Right $\gg$ All-Feet-Left
  $\ldots\sigma\sigma\sigma\sigma\sigma\sigma\ltimes \rightarrow \ldots\sigma\sigma\sigma(\sigma\sigma)\ltimes \rightarrow \ldots\sigma(\sigma\sigma)(\sigma\sigma)\ltimes \rightarrow \ldots$

Trochee $\gg$ Parse(σ)

- cannot parse hooves or iambs

Parse(σ) $\gg$ everything else

- everything gets parsed; worst things parse last

two more constraints

- *FootFoot:[17] AOVFE pair of adjacent feet.
- NonFinality:[18] AOVFE word-final syllable parsed into a foot.

[17] (Kager, 1994:8)

[18] (Prince and Smolensky, 1993/2004:51)

(37) Pseudo-nonfinality in Harmonic Serialism (unattested): the final syllable is passed over initially and then footed at the end of the derivation (Lamont, 2023)

/σσσσσσ/	Parse(σ)	Trochee	NonFinality	All-Feet-Right
a. $\sigma\sigma\sigma\sigma\sigma\sigma$	W 6			
b. $(\acute{\sigma}\sigma)\sigma\sigma\sigma\sigma$	4			W 4
c. $\sigma(\acute{\sigma}\sigma)\sigma\sigma\sigma$	4			W 3
d. $\sigma\sigma(\acute{\sigma}\sigma)\sigma\sigma$	4			W 2
☞e. $\sigma\sigma\sigma(\acute{\sigma}\sigma)\sigma$	4			1
f. $\sigma\sigma\sigma\sigma(\acute{\sigma}\sigma)$	4		W 1	L
g. $\sigma\sigma\sigma(\acute{\sigma}\sigma)\sigma$	W 4			L 1
h. $(\acute{\sigma}\sigma)\sigma(\acute{\sigma}\sigma)\sigma$	2			W 5
☞i. $\sigma(\acute{\sigma}\sigma)(\acute{\sigma}\sigma)\sigma$	2			4
j. $\sigma(\acute{\sigma}\sigma)(\acute{\sigma}\sigma)\sigma$	W 2	L		L 4
☞k. $(\acute{\sigma})(\acute{\sigma}\sigma)(\acute{\sigma}\sigma)\sigma$	1	1		9
l. $\sigma(\acute{\sigma}\sigma)(\acute{\sigma}\sigma)(\acute{\sigma})$	1	1	W 1	L 4
m. $(\acute{\sigma})(\acute{\sigma}\sigma)(\acute{\sigma}\sigma)\sigma$	W 1	L 1	L	9
💣 n. $(\acute{\sigma})(\acute{\sigma}\sigma)(\acute{\sigma}\sigma)(\acute{\sigma})$		2	1	9

(38) Pseudo-ternary rhythm in Harmonic Serialism (unattested): non-adjacent feet are parsed in the first pass (a-j), monosyllabic feet are parsed in the second pass (k-o) (Lamont, 2023)

/σσσσσσ/	PARSE(σ)	TROCHEE	*FOOTFOOT	ALL-FEET-LEFT
a. σσσσσσ	W 6			
☞ b. (σ́σ)σσσσ	4			
c. σ(σ́σ)σσσ	4			W 1
d. σσ(σ́σ)σσ	4			W 2
e. σσσ(σ́σ)σ	4			W 3
f. σσσσ(σ́σ)	4			W 4
g. (σ́σ)σσσσ	W 4			L
h. (σ́σ)(σ́σ)σσ	2		W 1	L 2
☞ i. (σ́σ)σ(σ́σ)σ	2			3
j. (σ́σ)σσ(σ́σ)	2			W 4
k. (σ́σ)σ(σ́σ)σ	W 2	L	L	L 3
l. (σ́σ)(σ́)(σ́σ)σ	2	1	W 2	L 5
💣 m. (σ́σ)σ(σ́σ)(σ́)	2	1	1	8
n. (σ́σ)σ(σ́σ)(σ́)	W 1	L 1	L 1	L 8
💣 o. (σ́σ)(σ́)(σ́σ)(σ́)		2	3	10

(39) Short word pathology (unattested): only disyllabic words are parsed into feet

		ALLFT-L	ALLFT-R	PARSE(σ)
a. /σσ/	i. σσ			W 2
	☞ ii. (σ́σ)			
b. /σσσ/	☞ i. σσσ			3
	ii. (σ́σ)σ		W 1	L 1
	iii. σ(σ́σ)	W 1		L 1

5.3 Directional constraints are good

Directional constraints assign vectors to candidates, not numbers (Eisner, 2000, 2002; Finley, 2008; Lamont, 2022)

(40) Violation vectors and harmonic ordering by PARSE$(\sigma)^{\Rightarrow}$

$$\underset{1\,2\,3\,4}{1111} > \underset{1\,2\,3\,4}{1110} > \underset{1\,2\,3\,4}{1101} > \underset{1\,2\,3\,4}{1100} > \underset{1\,2\,3\,4}{1011} > \underset{1\,2\,3\,4}{1001} > \underset{1\,2\,3\,4}{0111} > \underset{1\,2\,3\,4}{0011}$$

$$\underset{1}{\sigma}\underset{2}{\sigma}\underset{3}{\sigma}\underset{4}{\sigma} \prec \underset{1}{\sigma}\underset{2}{\sigma}\underset{3}{\sigma}(\underset{4}{\acute\sigma}) \prec \underset{1}{\sigma}\underset{2}{\sigma}(\underset{3}{\acute\sigma})\underset{4}{\sigma} \prec \left\{\begin{matrix}\underset{1}{\sigma}\underset{2}{\sigma}(\underset{3}{\acute\sigma}\underset{4}{\sigma})\\ \underset{1}{\sigma}\underset{2}{\sigma}(\underset{3}{\sigma}\underset{4}{\acute\sigma})\end{matrix}\right\} \prec \underset{1}{\sigma}(\underset{2}{\acute\sigma})\underset{3}{\sigma}\sigma \prec \left\{\begin{matrix}\underset{1}{\sigma}(\underset{2}{\acute\sigma}\underset{3}{\sigma})\underset{4}{\sigma}\\ \underset{1}{\sigma}(\underset{2}{\sigma}\underset{3}{\acute\sigma})\underset{4}{\sigma}\end{matrix}\right\} \prec (\underset{1}{\acute\sigma})\underset{2}{\sigma}\underset{3}{\sigma}\underset{4}{\sigma} \prec \left\{\begin{matrix}(\underset{1}{\acute\sigma}\underset{2}{\sigma})\underset{3}{\sigma}\underset{4}{\sigma}\\ (\underset{1}{\sigma}\underset{2}{\acute\sigma})\underset{3}{\sigma}\underset{4}{\sigma}\end{matrix}\right\}$$

(41) Violation vectors and harmonic ordering by Parse(σ)$^{\Leftarrow}$

$$\begin{array}{ccccccccccccccc}
1_4 1_3 1_2 1_1 & > & 1_4 1_3 1_2 0_1 & > & 1_4 1_3 0_2 1_1 & > & 1_4 1_3 0_2 0_1 & > & 1_4 0_3 1_2 1_1 & > & 1_4 0_3 0_2 1_1 & > & 0_4 1_3 1_2 1_1 & > & 0_4 0_3 1_2 1_1 \\
\sigma_1\sigma_2\sigma_3\sigma_4 & \prec & (\acute{\sigma}_1)\sigma_2\sigma_3\sigma_4 & \prec & \sigma_1(\acute{\sigma}_2)\sigma_3\sigma_4 & \prec & \left\{\begin{matrix}(\acute{\sigma}_1\sigma_2)\sigma_3\sigma_4 \\ (\sigma_1\acute{\sigma}_2)\sigma_3\sigma_4\end{matrix}\right\} & \prec & \sigma_1\sigma_2(\acute{\sigma}_3)\sigma_4 & \prec & \left\{\begin{matrix}\sigma_1(\acute{\sigma}_2\sigma_3)\sigma_4 \\ \sigma_1(\sigma_2\acute{\sigma}_3)\sigma_4\end{matrix}\right\} & \prec & \sigma_1\sigma_2\sigma_3(\acute{\sigma}_4) & \prec & \left\{\begin{matrix}\sigma_1\sigma_2(\acute{\sigma}_3\sigma_4) \\ \sigma_1\sigma_2(\sigma_3\acute{\sigma}_4)\end{matrix}\right\}
\end{array}$$

avoids the weird problems above: with Parse high ranked,
so too is the alignment

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